

Part A

Circuit A

1. Total Resistance = $20 + 30 + 50 = 100$
2. Total Current = $I = V/R = 125/100 = 1.25\text{A}$
3. Since Resistor in Series $I(1) = I(2) = I(3) = 1.25\text{A}$
4. Voltage drop across each resistor:
 $V_1 = 1.25 \times 20 = 25\text{ V}$
 $V_2 = 1.25 \times 30 = 37.5\text{ V}$
 $V_3 = 1.25 \times 50 = 62.5\text{ V}$
5. Power dissipated :
 $P_1 = 25 \times 1.25 = 31.25\text{ W}$
 $P_2 = 37.5 \times 1.25 = 46.875\text{ W}$
 $P_3 = 62.5 \times 1.25 = 78.125\text{ W}$
Total power = 156.25 W

Circuit B

1. Total Resistance:
 $1/R = 1/20 + 1/100 + 1/50 = 5/100 + 1/100 + 2/100 = 8/100$
 $R = 100/8 = 12.5\text{ Ohms}$
2. Total Current:
 $I = V / R = 125 / 12.5 = 10\text{ A}$
3. Current through each resistor:
 $I_1 (20\text{ Ohms}) = 125 / 20 = 6.25\text{ A}$
 $I_2 (100\text{ Ohms}) = 125 / 100 = 1.25\text{ A}$
 $I_3 (50\text{ Ohms}) = 125 / 50 = 2.5\text{ A}$
4. Voltage drop across each resistor:
 $V_1 = V_2 = V_3 = 125\text{ V}$

5. Power dissipated :

$$P_1 = 125 \times 6.25 = 781.25 \text{ W}$$

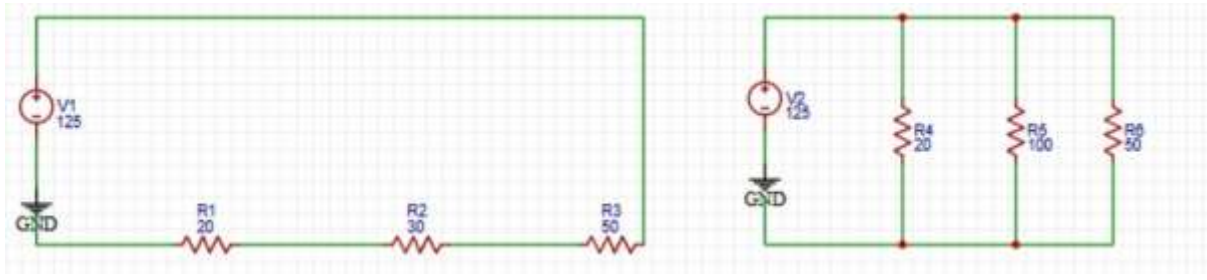
$$P_2 = 125 \times 1.25 = 156.25 \text{ W}$$

$$P_3 = 125 \times 2.5 = 312.5 \text{ W}$$

$$\text{Total power} = 1,250 \text{ W}$$

Part B

Question 1



Circuit a — Series Circuit

<u>Reference</u>	<u>Component</u>	<u>Value</u>	<u>Quantity</u>
<u>V1</u>	<u>DC Power Supply</u>	<u>125 V</u>	<u>1</u>
<u>R1</u>	<u>Resistor</u>	<u>20 Ω</u>	<u>1</u>
<u>R2</u>	<u>Resistor</u>	<u>30 Ω</u>	<u>1</u>
<u>R3</u>	<u>Resistor</u>	<u>50 Ω</u>	<u>1</u>

Circuit b — Parallel Circuit

<u>Reference</u>	<u>Component</u>	<u>Value</u>	<u>Quantity</u>
<u>V1</u>	<u>DC Power Supply</u>	<u>125 V</u>	<u>1</u>
<u>R1</u>	<u>Resistor</u>	<u>20 Ω</u>	<u>1</u>
<u>R2</u>	<u>Resistor</u>	<u>100 Ω</u>	<u>1</u>
<u>R3</u>	<u>Resistor</u>	<u>50 Ω</u>	<u>1</u>

Arduino Nano + DHT22

<u>Reference</u>	<u>Component</u>	<u>Value/Model</u>	<u>Quantity</u>
<u>U1</u>	<u>Microcontroller</u>	<u>Arduino Nano 33 IoT</u>	<u>1</u>

<u>Reference</u>	<u>Component</u>	<u>Value/Model</u>	<u>Quantity</u>
<u>U2</u>	<u>Temperature & Humidity Sensor</u>	<u>DHT22</u>	<u>1</u>
<u>R7</u>	<u>Resistor (Pull-up)</u>	<u>10 kΩ</u>	<u>1</u>
<u>W1</u>	<u>Connecting Wire</u>	<u>—</u>	<u>—</u>

Question2

